

RESPIRATORY SYSTEM

Definitions

Breathing - the pumping of air into and out of the lungs.

Respiration - the release of energy from food inside cells.

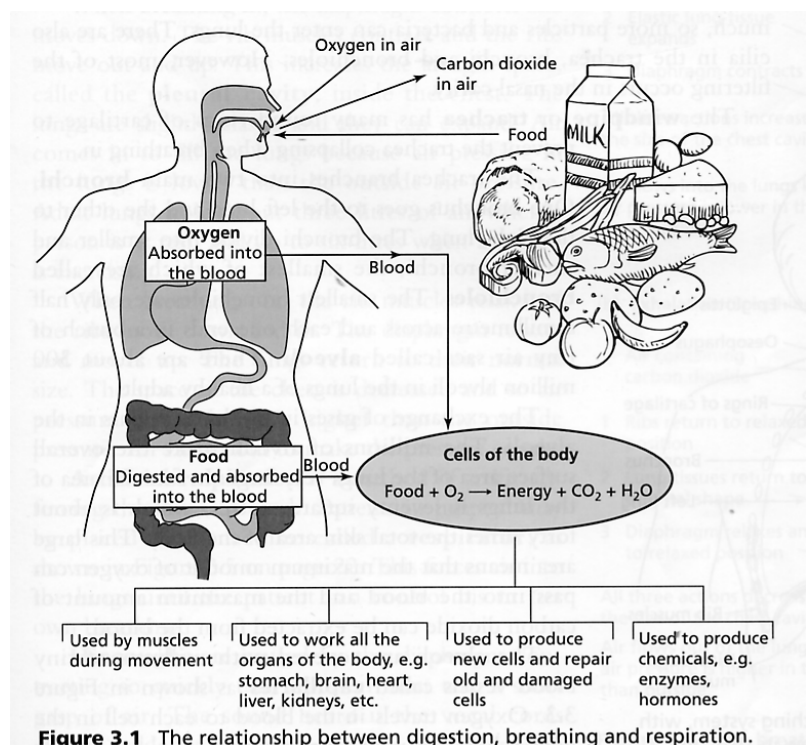
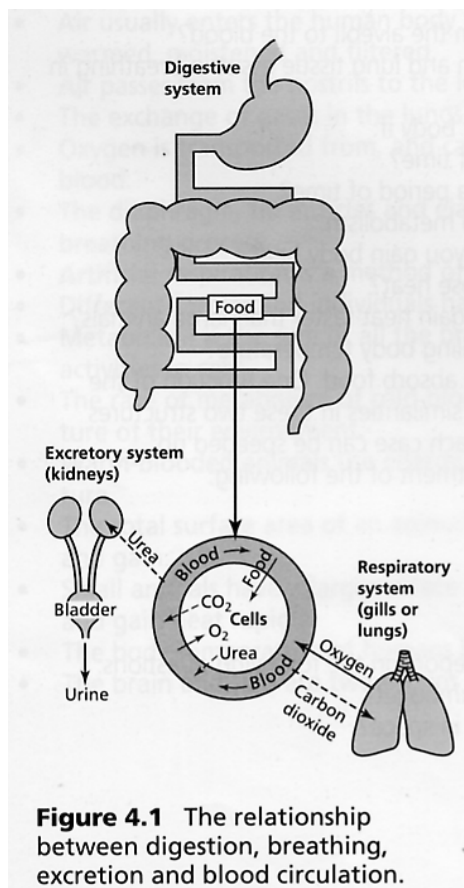
Respiration is the slow oxidation of food . It can be summarised as:



The food and O_2 are carried by the blood to each cell where respiration occurs.

The wastes of respiration (CO_2 and H_2O) are carried away from the cells by the blood and removed by the **kidneys**, **lungs** and **skin**.

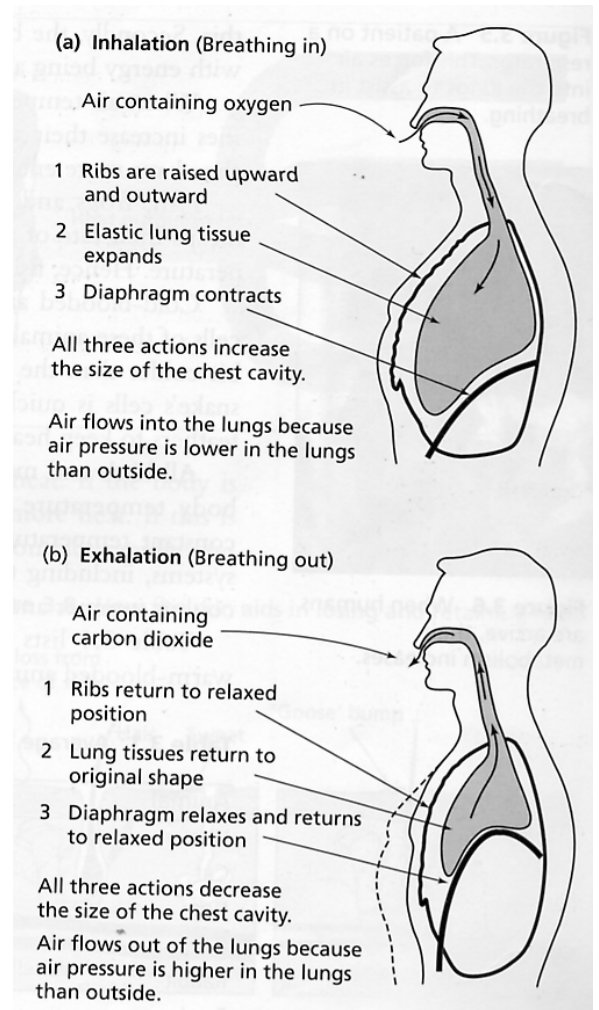
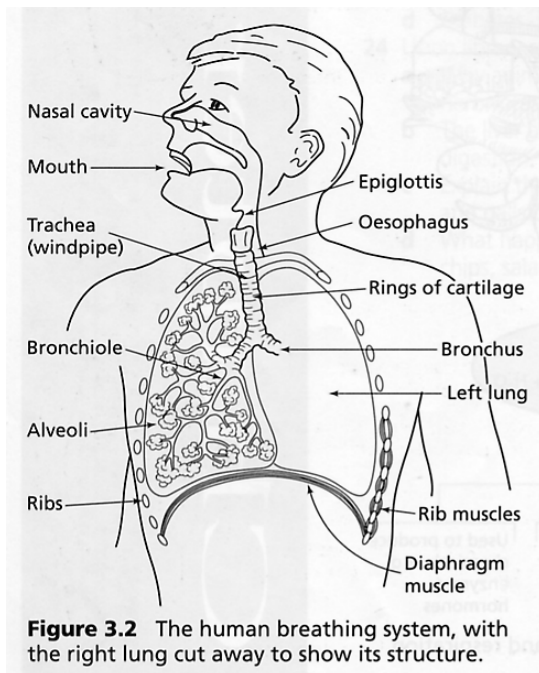
Thus, there is a connection between the digestive, breathing, circulatory and excretory system in the body.



Breathing occurs when muscles pull the ribs outwards and increases the volume of the lungs, causing a low pressure area.

Air at a higher pressure outside rushes in.

When the muscles relax, the volume of the lungs decreases creating a high pressure inside and the air rushes out.



Alveoli are small sacks inside the lungs which are surrounded by tiny blood vessels called **capillaries**.

It is here where O_2 is passed into the blood while CO_2 and H_2O are passed out into the lung. This is called **diffusion**.

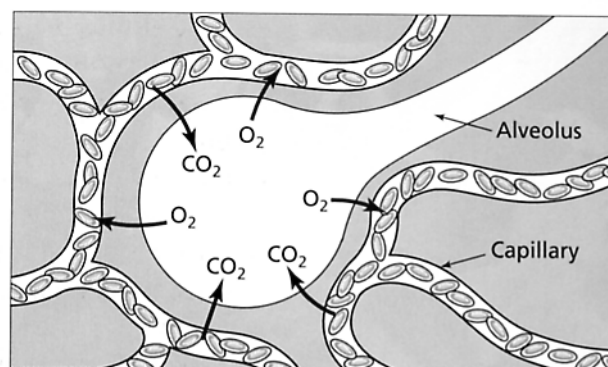


Figure 3.3 Gas exchange in an alveolus takes place between the thin, moist membrane of the alveolus and the capillary wall.